Summary of the Process for Determining M

1 Summary

This document summarizes the process for determining a fixed M.

Because the covariates originally considered for regression estimation of M are also likely to affect p, M is defined as the "total potential market." Accordingly, M_c is set as a fixed value (a city-specific constant) and calculated as city population × per-person trip generation. The data will be examined before the detailed calculation method is designed; only guidelines are presented below.

2 Previously Established Candidates

The following candidate driving factors have already been selected through data research in `B. Candidates/candidates_(M,p`

2.1 Driving factors of M

The papers below mainly explain **observed ride-hailing usage**, so they do not include the 'unrealized demand' included in the definition of M, making direct estimation impossible. However, I suppose the demand patterns provided by these papers lead to constructing 'potential travel opportunities M' through modeling.

Variable	Supporting paper	Scope established in the paper	Measurement data
Population and population density	Elkhouly and Alhadidi, *Ride-sharing Determinants* (paper); Zhang et al., *Examining spatial heterogeneity...* (paper)	Relationship of population and local density with conventional ride-hailing usage. Density effects may vary by area	ACS
Employment and economic activity	Elkhouly and Alhadidi (paper)	Uses the economically active population and employment environment to explain usage	LEHD/LODES, ACS
Income	Elkhouly and Alhadidi (paper)	Analyzes the relationship between income and usage in Chicago	ACS B19013
Share of car-free households	Elkhouly and Alhadidi (paper)	Analyzes the relationship between households without cars and usage	ACS B08201
Land-use mix, accessibility, and commercial activity	Zhang et al., *Examining spatial heterogeneity...* (paper)	Nonlinear and spatial differences in the urban environment and ridesourcing usage	EPA SLD
Restaurant and leisure-facility density	Loa, Lee, and Circella, *Just another ride-hailing?* (paper)	Relationship with **robotaxi trip counts in SF**. Does not separate the effects of M and p	Census CBP, OpenStreetMap
Public transport and walking environment	Zhang et al. (paper); Cats et al., *Beyond the dichotomy...* (paper)	Analyzes the local environment and travel-time competitiveness and substitutability with public transport. Not a uniform increase/decrease effect	EPA SLD, GTFS
Airport-related market size	Zhang et al. (paper)	Analyzes airport travel as a separate spatial category. **Does not estimate the direct effect of airport passenger counts**	FAA passenger data. Used as a proxy for market size
Time, day of week, and season	Elkhouly and Alhadidi (paper)	Analyzes differences in usage over time. Specific variables such as holidays are selected through separate design	Dates and times in trip data
Weather	Additional: Liu, Jiang, and Chen, *Quantifying the impact of weather on ride-hailing ridership: Evidence from Haikou, China* (2021, paper)	Analyzes the relationship between weather and ride-hailing usage. Effects in U.S. cities require revalidation	NOAA CDO

Note. All variables investigated using data such as those in Elkhouly & Alhadidi and Loa, Lee, & Circella concern districts within particular cities; an assumption of transfer across cities is therefore required.

3 Methods for Setting M_c as a Constant

Method	Calculation of M_c	Required data	Features	Selected?
(a) Reference city + assumed β	$\text{population}_c \times m_{\text{ref}} \times \prod_k \left(\frac{x_{c,k}}{x_{\text{ref},k}} \right)^{\beta_k}$	M for one reference city	β is assumed, so sensitivity analysis is essential	X
(b) Estimate β using district regression	Same as (a)	District-level data for the reference city	District-to-city transfer assumption required + β is assumed, so sensitivity analysis is essential	X
(c) Cross-classification of trip rates by population group	$\sum_g \text{population}_{c,g}$ × (per-person trip rate of group g)	Group-specific trip rates from NHTS (U.S. National Household Travel Survey) + city population composition from ACS	Standard trip-generation method in transport planning. Applicable to all cities using the same definition, without regression	O

4 If (a) or (b) Is Selected

For (a) and (b), estimate β, then calculate M_c once by substituting each candidate city's x_c and population into the equation.

However, M_c calculated in this way defines overall ride-hailing. Applying it to the previously defined potential market M_c requires a transfer assumption.

$$M_c = \text{population}_c \times m_{\text{ref}} \times \prod_k \left(\frac{x_{c,k}}{x_{\text{ref},k}} \right)^{\beta_k}$$

- m_{ref}: ride-hailing trips per person in the reference city (e.g., Chicago).
- β_k: percentage increase in per-person trips when variable k increases by 1% (elasticity).
- Methods for determining β:

Method	Description	Features
(a) Assume $\beta = 1$	Equal weights for all variables	Simple, but sensitivity experiments on β are essential
(b) Estimate from districts within the reference city	District trips in Chicago TNP data + district variables in ACS; regress $\log(\text{trips/population}) = a + \sum_k \beta_k \log x_k$	Many observations satisfy the $n/10$ rule of thumb. A district-to-city transfer assumption is required

4.1 Selection Criteria for the Driving Factors of M

No.	Criterion	Type
1	Literature reports an association with ride-hailing usage with a consistent sign; exclude or explain if the estimated sign differs	Before + after
2	Available at the decision time for all candidate cities from the same source, with the same definition and geographic unit	Before
3	Interpretable; number of variables $\leq n/10$; one per concept	Before
4	Scale-independent ratios (per person, area, or household); separate population as the scale factor	Before
5	Independent of decision q (exogenous)	Before
6	Candidate-city values lie within the training-data range	Before
7	After fitting, residuals exhibit no pattern with respect to x (zero mean, constant spread)	After

4.2 Discussion of Each Criterion

1. **Variables with a consistently signed association in the literature**
   - Meaning: exclude variables whose interpretation is ambiguously positive or negative, because interpretation becomes difficult. [Post-estimation validation] Even if the interpretation has a consistent direction, exclude or explain a fitted sign that differs from the literature.
   - Caution: the factors in the reviewed literature concern "observed ride-hailing usage," not the total potential market M. They cannot be declared to "definitely affect M." Describe selected variables as associated with ride-hailing usage and explicitly state transfer to M as an assumption.
2. **Variables with data available for all candidate cities**
   - (1) Same source and definition (e.g., ACS five-year estimates); (2) same geographic unit (city boundary vs. metropolitan area vs. service area must not be mixed); (3) values already available at the decision time (no future data).
3. **Interpretable variables**
   - Select variables that reveal "which context determines entry." Avoid selecting many correlated variables where possible (e.g., density, zero-vehicle households, public transport).
4. **Scale-independent ratios**
   - Density is per area, income is per household, and airport passengers are per person, so not all measures are "per person." Population itself is not a covariate but the scale factor in M = population $\times$ (per-person rate). The precise rule is that if the dependent variable is trips per person, the covariates should also be ratios.
5. **Exogenous variables**
   - Variables that do not change with decision q. For example, "past ride-hailing usage" generally passes on the M side, but using it as a covariate maximizes predictive power without explaining "why."
6. **Candidate-city values within the training range**
   - Outside the range, $\hat{f}(x_c)$ becomes extrapolation and the prediction error in Kannan's Assumption 2 increases.
7. **Covariates unrelated to residuals**
   - This is a post-fitting check. In OLS, correlation between included covariates and residuals is automatically zero, so that criterion cannot filter anything. The intended condition is Kannan's assumption that the residual distribution does not vary with x: zero mean and homoscedasticity. Check using residual plots or heteroscedasticity tests. If violated, transform the dependent variable into a ratio or log (the role of criterion 4) or add an omitted variable. Read this criterion as "check after selection: residuals must show no pattern with x; otherwise transform or add variables."

4.3 Applying the Criteria to the Candidates: M

4.4 Application Table

The following applies criteria 1–5 to each variable in the driving-factor table. Criteria 6 and 7 require data and regression and are therefore left for later. Decisions have three levels: select / defer (decide after checking data and signs) / exclude. A + under criterion 1 is the sign established in the paper; "check" indicates that no sign was recorded in the table and the original paper must be consulted again.

Variable	1	2	3	4	5	Decision	Reason / form of use
Population	–	◯	◯	×	◯	Separate	A scale factor rather than a covariate. Place it as the multiplier in M = population $\times \hat{f}(x)$ (criterion 4).

Population density	△	○	○	○	○	Defer	The table says "Density effects may vary by area." Check Zhang et al. to determine whether magnitude differs or the sign reverses. If both remain, examine correlation with zero-vehicle household share and reconsider under criterion 3 (one per concept).
Employment / economic activity	+	○	○	○	○	Select	Elkhouly & Alhadidi: economic activity share is significant and positive. LEHD/LODES and ACS cover all cities. Use jobs per person or economic activity share. Check correlation with income (3.3).
Income	+	○	○	○	○	Select	Elkhouly & Alhadidi: median income is significant and positive. ACS B19013 covers all cities. Household median income (a per-household value). Evidence is from one city (Chicago districts), so post-estimation city-level sign checks are essential.
Share of car-free households	+	○	○	○	○	Select	Elkhouly & Alhadidi: zero-vehicle household share is significant and positive. ACS B08201 covers all cities. Already a household share, naturally satisfying criterion 4. Most direct interpretation: no car implies ride-hailing demand.
Land-use mix / accessibility	Check	○	△	○	○	Defer	The table says "Nonlinear and spatial differences." Nonlinearity may still be monotonic (constant sign), so check Zhang et al. As an index, interpretation is indirect (criterion 3: △).
Restaurant / leisure density	△	○	○	○	○	Defer	Evidence concerns robotaxi trips in one city, SF, without separating M and p. CBP data are available. Reconsider as a p candidate.
Public transport / walking	×	○	○	○	○	Exclude	The table says "Not a uniform increase/decrease effect" (both substitution and complementarity). Fails criterion 1.
Airport-related market size	×	○	○	○	○	Defer	"Does not estimate the direct effect"; a proxy. Airport passengers per person could be used, but criterion 1 is not met. If included, label as a proxy and use only in sensitivity analysis.
Time / day of week / season	–	○	×	–	○	Exclude	Temporal rather than city characteristics. Averaged out in annual entry decisions. Consider only one year trend.
Weather	△	○	×	–	○	Exclude	Evidence is from Haikou, China, and needs U.S. validation (criterion 1 uncertain). As a temporal variable, averaged out in annual decisions.

4.5 Further Consideration: (1) Correlation between Variables (2) Unit of Analysis

(i) Correlation between economic activity share and income. The two are usually positively correlated but not identical: a university city may have high economic activity and low income, while a city with many retirees may show the reverse. The paper included both in the same spatial regression and reported both significant, suggesting distinct information at the Chicago district level. This must be checked again in our city-level data: inspect correlation coefficients and VIF; if correlation is high (e.g., $|r| > 0.7$ or VIF > 5), retain only one under criterion 3 (one per concept). Priority: zero-vehicle household share (most direct interpretation) > either income or economic activity share.

(ii) Transfer of the unit of analysis. The paper uses spatial regression of districts within one city, Chicago (community areas or census tracts: check the original paper). We intend to use city-level observations. Relationships between districts within a city are not guaranteed to hold between cities; aggregation may change the relationship. Label the criterion-1 evidence as "district-level evidence," assume city-level transfer explicitly, and supplement it with criterion-1 post-estimation validation (recheck signs using city-level data). Between-city income comparisons also incorporate cost-of-living differences, so consider regional price adjustment or relative values.

4.6 Preliminary Selection Results

- **Select (3, signs established):** economic activity share (or jobs per person), median household income, zero-vehicle household share. The dependent variable is ride-hailing trips per person, with population separated as the scale factor. Reduce a highly correlated pair to one variable (3.3).
- **Defer (4):** density; land-use mix (check sign reversal); restaurant/leisure-facility density (consider moving to p candidates); airport passengers (proxy, sensitivity analysis only).
- **Exclude (3):** public transport/walking environment; time/day of week/season; weather.

4.7 Conclusion on Methods (a) and (b)

- The selected covariates (economic activity share or jobs per person, median household income, and zero-vehicle household share) may be more likely to affect whether ride-hailing is chosen (p) than M.
- As discussed in 4.5, correlations between variables and the district-to-city transfer assumption require separate consideration.

5 If (c) Is Selected

M_c is constructed from NHTS group-specific per-person trip rates × city populations. This method produces a fixed value for every candidate city using the same definition, without regression.

The detailed calculation procedure will be revisited after examining the actual data. Only methodological guidelines are described below.

5.1 Required Data

Required item	Source (link)	Notes
Group-specific per-person trip rate r_g	NHTS (FHWA National Household Travel Survey) public microdata (household, person, trip files): nhts.ornl.gov/downloads	Respondents record a one-day travel diary. Includes household vehicle count, income bracket, household size, age, city-size categories, and travel modes. The 2017 survey is large, about 130,000 households including state add-ons; 2022 has a much smaller sample, requiring care when setting cell counts.
City population by group $N_{c,g}$	ACS five-year estimates (available for all geographic units): census.gov/programs-surveys/acs	For combinations of two or more variables absent from standard tables, aggregate ACS PUMS microdata directly (PUMA units with population $\geq$ 100,000).

Required item	Source (link)	Notes
City population (scale factor)	ACS B01003 or Census	
Local household travel survey for validation	MPO data, e.g., Chicago CMAP cmap.illinois.gov, New York NYMTC nymtc.org	Compare total trips to assess error size.

Note. Links are current as of writing. Change the year suffix of table identifiers (e.g., ACSDT5Y2022 for B08201) to match the ACS year used.

Note. ACS table search: data.census.gov.

Note. ACS table examples: B08201 (household size × vehicles available), B19001 (household income brackets), B01001 (age).

Note. Cross-tabulation through ACS PUMS: census.gov/programs-surveys/acs/microdata.

Note. Aggregation tool: data.census.gov/mdat.

Note. Both are public datasets applicable to all candidate cities using the same definition, satisfying criterion 2 (same source, definition, and geographic unit). However, public NHTS files do not directly identify cities (only states, city-size categories, and some metropolitan-area codes). Thus the method multiplies "national group-specific trip rates," rather than "city-specific trip rates," by each city's population composition. This is the core of cross-classification.

5.2 Procedure

1. **Define groups g.** Household vehicle count and income best distinguish trip generation in transport planning and overlap with the initial covariate selection for M_c.
2. **Calculate r_g from NHTS.** Divide the trip file by group and use weights to calculate annual trips per person for group g.
3. **Obtain $N_{c,g}$ from ACS.** Obtain group populations for each candidate city. Vehicle count × income is absent from standard tables, so aggregate by PUMA from PUMS and combine into cities. If records are household-level, multiply by household size to obtain person counts.
4. **Multiply and sum.** All candidate cities use the same r_g, so differences between cities arise only from population size and composition.
$$M_c = \sum_{g \in G} N_{c,g} r_g.$$
Separating population as the scale factor gives
$$M_c = \text{population}_c \times \sum_{g \in G} \pi_{c,g} r_g, \qquad \pi_{c,g} = \frac{N_{c,g}}{\text{population}_c} \quad \text{(group share in city } c\text{)}.$$
Thus $\bar{r}_c = \sum_g \pi_{c,g} r_g$ is city c's "composition-weighted per-person trip rate."
5. **Validate.** Compare total trips with local MPO household travel surveys where available (e.g., Chicago CMAP, New York NYMTC) to assess error size, then use that size as the sensitivity range for M.
$$e_c = \frac{M_c - M_c^{\text{MPO}}}{M_c^{\text{MPO}}}, \qquad \text{sensitivity range: } M_c \times \left(1 \pm \max_{c'} |e_{c'}|\right).$$

5.3 Example

Group g (vehicles × income)	r_g (annual trips per person)	City A population $N_{A,g}$	$N_{A,g} \times r_g$
0 vehicles × low income	900	60,000	54.0M
0 vehicles × middle/high income	1,000	40,000	40.0M
1 vehicle × low income	1,100	150,000	165.0M
1 vehicle × middle/high income	1,250	250,000	312.5M
2+ vehicles × all	1,350	500,000	675.0M

Group g (vehicles × income)	r_g (annual trips per person)	City A population $N_{A,g}$	$N_{A,g} \times r_g$
Total		1,000,000	$M_A =$ 1,246.5M

5.4 Definitions to Establish in Advance

- **Which trips count as "usage opportunities"?** Including all person trips makes M very large and p very small. A trip such as a 300 m walk is not an opportunity to substitute a robotaxi, so one option is to restrict the definition to, for example, "non-walking trips beyond a certain distance." NHTS includes trip distance and mode, allowing this filter.
- **Geographic unit.** A robotaxi service area is part of a metropolitan area rather than an entire city. Fix one unit among city (place) boundaries, counties, and CBSAs, and extract ACS at the same unit (criterion 2).
- **Year.** The unit period is set to one year. Examine data from the same year and ensure an adequate sample size.

5.5 Limitations

- Group-specific trip rates are assumed nationally common, so city characteristics such as density and public transport are reflected only through population composition. To add further city effects, combine with (c) district regression.
- Small NHTS samples per cell make r_g unstable. Options include reducing the number of cells or using the larger 2017 sample.
- The observed p must be measured as "robotaxi trips/M_c" within the same geographic unit to align denominators. City-specific robotaxi trip-count data require separate verification.